PART A — QUESTIONS

AF9-001
Question:
Solve: $11x + 5 = 71$
A) 5
B) 6
C) 7
D) 8

AF9-002
Question:
If $9n - 4 = 59$, what is n?
A) 6
B) 7
C) 8
D) 9

AF9-003
Question:
Simplify: $6(2x + 3) - 5x$
A) $7x + 18$
B) $12x + 18$
C) $7x + 3$
D) $17x + 18$

AF9-004
Question:
A company charges $13 per unit plus a $40 processing fee. Which expression represents the total charge for u units?
A) $13u + 40$
B) $40u + 13$
C) $53u$
D) $40 - 13u$

AF9-005
Question:
If $f(x) = 5x + 6$, what is $f(8)$?
A) 34
B) 40
C) 46
D) 54

AF9-006
Question:
Which ordered pair satisfies the equation $y = 4x + 1$?
A) (2, 8)
B) (3, 13)
C) (4, 18)
D) (5, 22)

AF9-007
Question:
Solve: 10(x - 2) = 60
A) 6
B) 7
C) 8
D) 9

AF9-008
Question:
A pattern begins: 8, 17, 26, 35, ... What is the next number?
A) 42
B) 43
C) 44
D) 45

AF9-009
Question:
Solve: x/15 = 4
A) 19
B) 45
C) 60
D) 75

AF9-010
Question:
A line has a slope of -5 and crosses the y-axis at 11. Which equation represents the line?
A) y = 11x - 5
B) y = -5x + 11
C) y = 5x + 11
D) y = -11x + 5

AF9-011
Question:
If 6 cartons contain 108 parts, how many parts are in 4 cartons at the same rate?
A) 54
B) 60
C) 72
D) 84

AF9-012
Question:
Solve: 50 - 8x = 18
A) 3
B) 4
C) 5
D) 6

AF9-013
Question:
Which expression is equivalent to 27x - 12x + 4?

A) 15x + 4
B) 39x + 4
C) 15x - 4
D) 27x - 8

AF9-014
Question:
A tank starts with 140 gallons and drains 12 gallons each minute. Which equation gives the amount A left after m minutes?
A) A = 140m - 12
B) A = 12m + 140
C) A = 140 - 12m
D) A = 12m - 140

AF9-015
Question:
If y = -8x + 50, what is y when x = 5?
A) 10
B) 15
C) 40
D) 90

AF9-016
Question:
Solve: 14x - 6 = 8x + 42
A) 6
B) 7
C) 8
D) 9

AF9-017
Question:
Which value of x makes 7x + 2 greater than 44?
A) 5
B) 6
C) 7
D) 4

AF9-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: -3, 6, 15, 24

What is the rule?
A) y = 9x - 3
B) y = 3x + 9
C) y = -3x + 9
D) y = 9x + 3

AF9-019
Question:
Solve: $6(x + 4) = 3x + 45$
A) 5
B) 6
C) 7
D) 8

AF9-020
Question:
If $a = -9$ and $b = 4$, what is $a + 5b$?
A) -29
B) -11
C) 11
D) 29

AF9-021
Question:
Which point is located on the y-axis?
A) (7, 0)
B) (0, -8)
C) (-5, 3)
D) (4, -4)

AF9-022
Question:
A worker completes 96 checks in 12 minutes. At the same rate, how many checks can the worker complete in 7 minutes?
A) 48
B) 54
C) 56
D) 64

AF9-023
Question:
Solve for k: $S = 10k - 30$
A) $k = S + 30$
B) $k = (S + 30)/10$
C) $k = 10(S + 30)$
D) $k = S/10 - 30$

AF9-024
Question:
What is the slope of a line that goes through (4, 3) and (10, 21)?
A) 2
B) 3
C) 4
D) 18

AF9-025
Question:

Simplify: $19x - 8 - 11x + 14$
A) $8x + 6$
B) $30x + 6$
C) $8x - 22$
D) $30x - 22$

AF9-026
Question:
If $g(x) = x^2 - 5x$, what is $g(9)$?
A) 36
B) 45
C) 54
D) 81

AF9-027
Question:
A sequence follows the rule “multiply by 4, then subtract 3.” If the first term is 5, what is the second term?
A) 17
B) 20
C) 23
D) 27

AF9-028
Question:
Solve: $9(x - 1) + 2 = 65$
A) 6
B) 7
C) 8
D) 9

AF9-029
Question:
A line crosses the y-axis at 10 and falls 6 units for every 1 unit to the right. Which equation represents the line?
A) $y = 6x + 10$
B) $y = -6x + 10$
C) $y = 10x - 6$
D) $y = -10x + 6$

AF9-030
Question:
If $11/14 = x/42$, what is x?
A) 22
B) 27
C) 30
D) 33

AF9-031
Question:
Solve: $-11x + 18 = -70$

A) -8
B) -6
C) 6
D) 8

AF9-032
Question:
A box starts with 720 labels. Each shipment uses 40 labels. Which expression gives the number of labels left after s shipments?
A) 720 + 40s
B) 720 - 40s
C) 40s - 720
D) 720 ÷ 40s

AF9-033
Question:
Which equation has x = 14 as its solution?
A) 2x + 5 = 31
B) 3x - 8 = 34
C) x/7 + 6 = 9
D) 4x - 10 = 42

PART B — ANSWER KEY + EXPLANATIONS

AF9-001 — Correct: B — Explanation:
Solve 11x + 5 = 71. Subtract 5 from both sides: 11x = 66. Divide by 11: x = 6. The most common mistake is subtracting 5 correctly but forgetting to divide by 11.

AF9-002 — Correct: B — Explanation:
Start with 9n - 4 = 59. Add 4 to both sides: 9n = 63. Divide by 9: n = 7. A common mistake is subtracting 4 instead of adding it.

AF9-003 — Correct: A — Explanation:
Distribute first: 6(2x + 3) = 12x + 18. Then subtract 5x: 12x - 5x + 18 = 7x + 18. A common mistake is forgetting to multiply 3 by 6.

AF9-004 — Correct: A — Explanation:
The company charges $13 for each unit, so that part is 13u. The $40 processing fee is added one time. The total charge is 13u + 40. A common mistake is treating the processing fee as the per-unit charge.

AF9-005 — Correct: C — Explanation:
Substitute 8 for x: f(8) = 5(8) + 6 = 40 + 6 = 46. A common mistake is multiplying correctly but forgetting to add 6.

AF9-006 — Correct: B — Explanation:
Test the choices in y = 4x + 1. For (3, 13), 4(3) + 1 = 12 + 1 = 13, which matches the y-value. A common mistake is choosing a point that is close but not exact.

AF9-007 — Correct: C — Explanation:
Solve $10(x - 2) = 60$. Divide both sides by 10: $x - 2 = 6$. Add 2: $x = 8$. A common mistake is adding 2 before undoing the multiplication by 10.

AF9-008 — Correct: C — Explanation:
The pattern increases by 9 each time: 8, 17, 26, 35. Add 9 to 35: 44. A common mistake is assuming the pattern increases by 8 because the first number is 8.

AF9-009 — Correct: C — Explanation:
$x/15 = 4$ means x divided by 15 equals 4. Multiply both sides by 15: $x = 60$. A common mistake is adding 15 and 4 instead of multiplying.

AF9-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -5 and the y-intercept is 11, so the equation is $y = -5x + 11$. A common mistake is switching the slope and y-intercept.

AF9-011 — Correct: C — Explanation:
Find the unit rate: 108 parts ÷ 6 cartons = 18 parts per carton. For 4 cartons: $18 \times 4 = 72$. A common mistake is multiplying 108 by 4 without finding the rate first.

AF9-012 — Correct: B — Explanation:
Solve $50 - 8x = 18$. Subtract 50 from both sides: $-8x = -32$. Divide by -8: $x = 4$. A common mistake is losing the negative sign after subtracting 50.

AF9-013 — Correct: A — Explanation:
Combine like terms: $27x - 12x = 15x$. The constant 4 stays the same. The expression becomes $15x + 4$. A common mistake is combining the constant with the variable terms.

AF9-014 — Correct: C — Explanation:
The tank starts with 140 gallons and loses 12 gallons each minute. After m minutes, 12m gallons have drained. The amount left is $A = 140 - 12m$. A common mistake is adding 12m even though the tank is draining.

AF9-015 — Correct: A — Explanation:
Substitute $x = 5$ into $y = -8x + 50$: $y = -8(5) + 50 = -40 + 50 = 10$. A common mistake is treating $-8(5)$ as positive 40.

AF9-016 — Correct: C — Explanation:
Solve $14x - 6 = 8x + 42$. Subtract 8x: $6x - 6 = 42$. Add 6: $6x = 48$. Divide by 6: $x = 8$. A common mistake is adding 6 to only one side.

AF9-017 — Correct: C — Explanation:
Solve $7x + 2 > 44$. Subtract 2: $7x > 42$. Divide by 7: $x > 6$. The only listed value greater than 6 is 7. A common mistake is choosing 6, but $7(6) + 2 = 44$, not greater than 44.

AF9-018 — Correct: A — Explanation:
The y-values increase by 9 each time x increases by 1, so the slope is 9. When $x = 0$, $y = -3$, so the y-intercept is -3. The rule is $y = 9x - 3$. A common mistake is using +3 instead of -3.

AF9-019 — Correct: C — Explanation:
Distribute: $6x + 24 = 3x + 45$. Subtract $3x$: $3x + 24 = 45$. Subtract 24: $3x = 21$. Divide by 3: $x = 7$. A common mistake is forgetting to multiply 4 by 6.

AF9-020 — Correct: C — Explanation:
Substitute $a = -9$ and $b = 4$: $a + 5b = -9 + 5(4) = -9 + 20 = 11$. A common mistake is treating -9 as positive 9.

AF9-021 — Correct: B — Explanation:
A point on the y-axis has $x = 0$. The point (0, -8) is on the y-axis. A common mistake is choosing (7, 0), which is on the x-axis.

AF9-022 — Correct: C — Explanation:
96 checks in 12 minutes means $96 \div 12 = 8$ checks per minute. In 7 minutes: $8 \times 7 = 56$. A common mistake is multiplying 96 by 7 without finding the rate first.

AF9-023 — Correct: B — Explanation:
Start with $S = 10k - 30$. Add 30 to both sides: $S + 30 = 10k$. Divide by 10: $k = (S + 30)/10$. A common mistake is dividing S by 10 first and then adding 30.

AF9-024 — Correct: B — Explanation:
Slope equals change in y divided by change in x. From (4, 3) to (10, 21), y increases by 18 and x increases by 6. The slope is $18 \div 6 = 3$. A common mistake is using 18 as the slope without dividing by the change in x.

AF9-025 — Correct: A — Explanation:
Combine like terms: $19x - 11x = 8x$, and $-8 + 14 = 6$. The simplified expression is $8x + 6$. A common mistake is writing $8x - 6$ because the positive 14 is missed.

AF9-026 — Correct: A — Explanation:
Substitute 9 for x: $g(9) = 9^2 - 5(9) = 81 - 45 = 36$. A common mistake is calculating 9^2 as 18 instead of 81.

AF9-027 — Correct: A — Explanation:
Start with 5. Multiply by 4: 20. Then subtract 3: 17. A common mistake is subtracting 3 first and then multiplying.

AF9-028 — Correct: C — Explanation:
Solve $9(x - 1) + 2 = 65$. Distribute: $9x - 9 + 2 = 65$. Combine: $9x - 7 = 65$. Add 7: $9x = 72$. Divide by 9: $x = 8$. A common mistake is forgetting to combine -9 and +2.

AF9-029 — Correct: B — Explanation:
“Falls 6 units for every 1 unit to the right” means the slope is -6. The line crosses the y-axis at 10, so the equation is $y = -6x + 10$. A common mistake is using a positive slope because the number 6 is given.

AF9-030 — Correct: D — Explanation:
Use the proportion $11/14 = x/42$. Since $14 \times 3 = 42$, multiply 11 by 3: $x = 33$. A common mistake is adding 28 to the numerator because 14 becomes 42.

AF9-031 — Correct: D — Explanation:
Solve $-11x + 18 = -70$. Subtract 18 from both sides: $-11x = -88$. Divide by -11: $x = 8$. A common mistake is giving -8 because both sides contain negative numbers.

AF9-032 — Correct: B — Explanation:
The box starts with 720 labels. Each shipment uses 40 labels, so s shipments use 40s labels. The number left is $720 - 40s$. A common mistake is adding 40s even though labels are being used.

AF9-033 — Correct: B — Explanation:
Test $x = 14$. In choice B, $3(14) - 8 = 42 - 8 = 34$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.